

Classroom



In this section of *Resonance*, we invite readers to pose questions likely to be raised in a classroom situation. We may suggest strategies for dealing with them, or invite responses, or both. “Classroom” is equally a forum for raising broader issues and sharing personal experiences and viewpoints on matters related to teaching and learning science.

Ranking Events Based on a Random Sample*

Quyen Tran
DePauw University
Greencastle, Indiana, USA
quyentran.2027@depauw.edu

Jyotirmoy Sarkar
Indiana University Indianapolis
Indianapolis, Indiana, USA
jsarkar@iu.edu

When a random sample of items is classified into categories, which are then ranked from the most frequent to the least, a decision made too early is prone to being wrong. How large must a sample be so that we are reasonably assured of a correct ranking?

1. Two Examples of Ranking

A biology class went on a field trip to determine the relative abundance of male and female *Nemipterus furcosus* population in the Pahang coastal water, Malaysia. See [1]. In one hour, they caught 51 *N. furcosus* fish of which 29 were female. They concluded that females were more abundant than males, except that Kamal raised a concern: “What if we waited until we caught four or five times as many fish? Could our decision be flipped?”

A geology club classified 245 rock samples into five mineral types: (a) K-feldspar, (b) Plagioclase feldspar, (c) Quartz, (d) Biotite, and (e) Hornblende. See [2]. If the frequencies are: (a) 36, (b)

A decision made too early may be wrong.

Frequencies are necessary for ranking, but may not be sufficient for accuracy.

*Vol.31, No.2, DOI: <https://doi.org/10.1007/s12045-026-1943-1>



Keywords

Probability mass function,
cumulative distribution function,
ranking, recurrence relation,
sample size determination.

Probability is applied in
many sciences.

66, (c) 43, (d) 52, (e) 48, can the club rank the five mineral types as $a < c < e < d < b$? Or must they classify many more rock samples before determining the ranking?

We study probability. Let us formulate a problem whose solution will enlighten both groups of science students.

Problem. Suppose that a random phenomenon can result in one of three mutually exclusive and exhaustive events— A, B, C —with respective **unknown** probabilities $p, q, r \in (0, 1)$ such that $p + q + r = 1$ and no two probabilities are equal. How large a sample must you take, and how should you rank the three events as the most likely, the medium likely, and the least likely?

2. Ranking Events by Sorting Frequencies

Suppose that each execution of the random phenomenon costs us ₹1. Let $X_1, X_2, X_3, \dots, X_n$ be an independent and identically distributed sequence of n outcomes of this random phenomenon, obtained by spending ₹ n . Let the frequencies of the three events be N_A, N_B, N_C such that $N_A + N_B + N_C = n$.

Since n is non-random,
the last frequency is
known, given the others.

As n grows larger, the
unknown proportions are
estimated more
accurately.

The strong law of large numbers (see, [3] and, for example, [4]) declares that with probability one, $(N_A, N_B, N_C)/n$ converges to (p, q, r) as $n \rightarrow \infty$. Moreover, (N_A, N_B, N_C) is a *sufficient statistic* for (p, q, r) ; that is, the entire random sample of n outcomes contains absolutely no more information about (p, q, r) other than through (N_A, N_B, N_C) ! See [4]. Therefore, it is sensible to rank the three events by sorting $\{N_A, N_B, N_C\}$ in decreasing order.

If n is small, the ranking
may be wrong.

Minimize resource
investment, but ensure
high enough accuracy.

However, problems arise when, because of resource limitation, the sample size n is small: There may be ties among $\{N_A, N_B, N_C\}$, in which case, we may break ties using extraneous randomness such as rolling a fair die, and therefore risk incorrect ranking. Or with some non-negligible probability, $\{N_A, N_B, N_C\}$ may not be ordered the same way as the corresponding events. Or both.

The objective of this paper is to solve the interrelated problems: (1) Given n , what is the probability of correctly ranking the events as a function of (p, q, r) ? (2) Given (p, q, r) , what is the small-



est sample size n that guarantees a high enough probability (say, 95%) of correctly ranking the events?

3. Dichotomous Random Phenomenon

Before solving the three-events problem stated above, let us recall the solution to the special case where the random phenomenon results in one of two events— $A, B = A^c$. That is, in this special case, $r = 0$. Solving a special case is often fruitful. Assume that $p \neq q$, or $p \neq 1/2$. The problem simplifies to using a random sample of n outcomes to determine whether $p > q$ or $p < q$; or equivalently, whether $p > 1/2$ or $p < 1/2$.

For example, are female *N. furcosus* more abundant than male?

Here, $N_A = N_A(n)$ denotes the frequency of event A among n independent executions of the random phenomenon, and it follows a binomial(n, p) distribution; and $N_B(n) = n - N_A(n)$. In particular, the probability mass function (PMF) of $N_A(n)$ is given by

N_A follows a binomial(n, p) distribution.

$$P\{N_A(n) = x\} = \binom{n}{x} p^x q^{n-x}, \text{ for } x = 0, 1, 2, \dots, n \quad (1)$$

where $\binom{n}{x} = n!/[x!(n-x)!]$ is called the binomial coefficient. By conditioning on the first (or the last) outcome, the PMF in (1) can be computed recursively as

$$P\{N_A(n) = x\} = q P\{N_A(n-1) = x\} + p P\{N_A(n-1) = x-1\}.$$

Equivalently, the binomial coefficients satisfy the recursive relations (see [5])

Recursive relations provide an easy way to compute step-by-step.

$$\binom{n}{x} = \binom{n-1}{x} + \binom{n-1}{x-1} \quad (2)$$

and can be arranged as rows to form Pascal's triangle (see [6]).

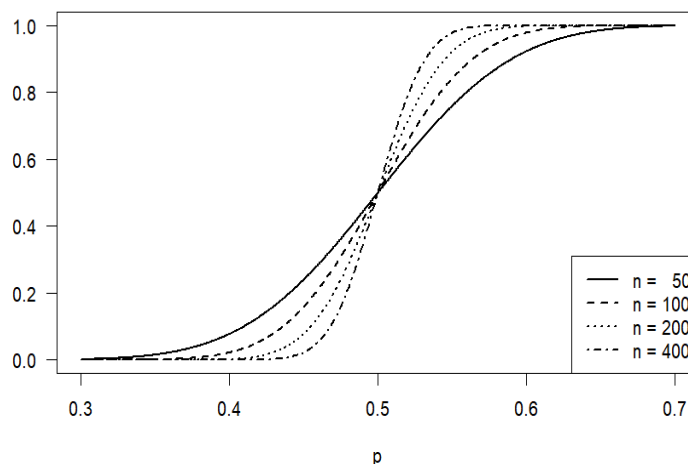
Based on a sample of size n , the probability of concluding that " $p > q$ " is given by

Binomial CDF and PMF determine $P("p > q")$.

$$\begin{aligned} P("p > q") &= P\{N_A > N_B\} + \frac{1}{2} P\{N_A = N_B\} \\ &= P\{N_A > n/2\} + \frac{1}{2} P\{N_A = n/2\} \\ &= 1 - P\{N_A \leq n/2\} + \frac{1}{2} P\{N_A = n/2\}. \end{aligned} \quad (3)$$



Figure 1. The probability of concluding “ $p > q$ ”, as a function of $p \in (0, 1)$, for sample sizes $n = 50, 100, 200, 400$.



That is, one minus the cumulative distribution function (CDF), plus half the PMF of a binomial(n, p) variable evaluated at $n/2$. Since N_A must be an integer, the PMF is 0 when n is odd.

Figure 1, shows the probability of declaring “ $p > 1/2$ ” as a function of $p \in (0, 1)$ for different sample sizes. The R codes are given in the *Appendix*.

If p is higher, we are more likely to conclude “ $p > q$ ”.

The graph of the function exhibits a 180° rotation symmetry.

The biology class would do well to pay heed to Kamal’s concern.

For each n , the probability $P(“p > q”)$, of concluding “ $p > q$ ”, is a monotonically increasing function of $p \in (0, 1)$. For each fixed $p > 1/2$, $P(“p > q”)$ increases with n . Similarly, for each fixed $p < 1/2$, $P(“p > q”)$ decreases as n increases. In fact, by symmetry, $P(“p > q”|p) = P(“p < q”|1 - p)$, which by the complement rule equals $1 - P(“p > q”|1 - p)$. When $p = 1/2$, irrespective of the sample size n , it is a pure gamble whether we conclude $p > q$ or $p < q$.

Suppose that the true proportion of male *N. furcosus* fish is $p = .55$, then based on a sample of $n = 51$, the biology class could correctly conclude that “male fish are more abundant” with probability $1 - \text{pbinom}(25, 51, .55) = .7641$. If $p = .52$, the probability



n	100	150	200	250	300
$P("p > q")$	.8413	.8899	.9216	.9433	.9586
n	255	265	275	285	295
$P("p > q")$	.9455	.9488	.9519	.9547	.9576
n	266	267	268	269	270
$P("p > q")$	.9488	.9495	.9495	.9501	.9501

Table 1. The probability of correctly concluding that " $p > q$ " when $p = .55$ for various values of n .

of such a correct conclusion is $1 - \text{pbinom}(25, 51, .52) = .6130$. Thus, Kamal had raised a concern worth pondering over.

Conversely, when the true value is $p = .55$ (hence, $q = .45$), how large a sample must we take so that with probability at least 95% we will make the correct conclusion that " $p > q$ "?

The computations in *Table 1*, justify that the requisite n must be at least 269. First, we search for n over a wide interval with large increments and then successively zoom into narrower intervals with smaller increments. See R codes in the *Appendix*.

We leave it to the reader to verify that the answer to the same question when $p = .52$ is at least 1691. The reader should also check that the answer is always an odd number because

$$P("p > q" | n = 2m - 1) = P("p > q" | n = 2m).$$

First, search broadly, then zoom in.

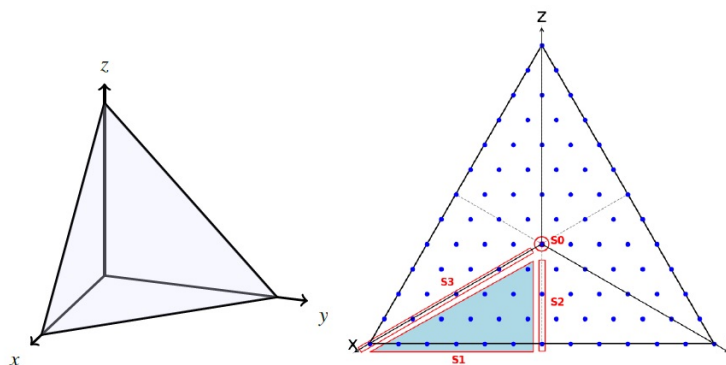
If $N_A + N_B = n$ is odd, $N_A \neq N_B$; that is, a tie is impossible.

That is, if n is even, skip the last execution of the random process (and save ₹1) without losing the precision to predict correctly!

Admittedly, the above calculations are computationally intensive. An alternative is to use the central limit theorem, according to which, for large n , N_A is approximately normally distributed with mean np and variance $np(1 - p)$. See [4] or [7]. Therefore, to ensure a 95% probability of correctly concluding that " $p > q$ " choose n to satisfy $P(N_A < n/2) = .05$. Equivalently, $n/2 \approx np + z\sqrt{np(1 - p)}$, where $z = \text{qnorm}(.05, 0, 1) = -1.645$ is the fifth percentile of a standard normal distribution. After simplifying, we have $n \approx z^2 p(1 - p) / (p - 1/2)^2$. According to this approximate formula, $n = 268$ when $p = .55$, and $n = 1689$ when $p = .52$.

The larger the n , the better the binomial distribution is approximated by the normal distribution.

Figure 2. Visualizing the support set $\mathcal{S}$ and its partition into subsets for computing the necessary probabilities.



(a) Support set $\mathcal{S}$ in 3D

(b) Partition of $\mathcal{S}$ for $n = 12$

4. Trichotomous Random Phenomenon

$\mathcal{S}$ is called a 2D lattice.

In the ranking problem of a trichotomous random phenomenon, (N_A, N_B, N_C) has a multinomial($n, (p, q, r)$) distribution with PMF on the support set $\mathcal{S} = \{0 \leq x, y, z \leq n; x + y + z = n\}$ given by

$$f_n(x, y, z) = P\{N_A(n) = x, N_B(n) = y, N_C(n) = z\} = \binom{n}{x, y, z} p^x q^y r^z, \quad (4)$$

Readers, please do this small exercise.

S_0 : the central singleton (circled point), where $\{N_A = N_B = N_C\}$ (occurs only when 3 divides n);
 S_1 : the triangular region, where $\{N_A > N_B > N_C\}$;
 S_2 : the vertical line (along the z -axis), where $\{N_A = N_B > N_C\}$;
 S_3 : the slant line (along the x -axis), where $\{N_A > N_B = N_C\}$.

where $\binom{n}{x, y, z} = n!/[x!y!z!]$ is called the trinomial coefficient. The reader can check that $\mathcal{S}$ has $\binom{n+2}{2}$ elements.

Figure 2(a) shows $\mathcal{S}$ in 3D with respect to the (x, y, z) coordinates. Figure 2(b) shows all elements of $\mathcal{S}$ for $n = 12$ and partitions them into 13 subsets S_0, S_1, S_2, S_3 , etc.

Readers, please count the number of elements in subsets S_1, S_2, S_3 , etc. for any n .

As in the previous section, by conditioning on the first (or the last) outcome, the PMF in (4) is computed recursively as

$$f_n(x, y, z) = r f_{n-1}(x, y, z-1) + q f_{n-1}(x, y-1, z) + p f_{n-1}(x-1, y, z)$$

Equivalently, the trinomial coefficients satisfy

Recursive relation has recurred in this paper!

$$\binom{n}{x, y, z} = \binom{n-1}{x, y, z-1} + \binom{n-1}{x, y-1, z} + \binom{n-1}{x-1, y, z} \quad (5)$$

and can be arranged as layers to form a Pascal's pyramid (see [6]). Readers, please build this pyramid.

You can brag about building a pyramid!

Since no two elements of $\{p, q, r\}$ are equal, based on a sample of size n , we can conclude one of the following six possibilities

$$\begin{aligned} C1 : (p > q > r), \quad C2 : (p > r > q), \quad C3 : (q > p > r), \\ C4 : (q > r > p), \quad C5 : (r > p > q), \quad C6 : (r > q > p), \end{aligned}$$

with associated probability

Recall ties are broken by rolling a fair die.

$$\begin{aligned} P(C1) &= h(p, q, r), \text{ say} \\ &= P\{N_A > N_B > N_C\} + \frac{1}{2}P\{N_A = N_B > N_C\} \\ &\quad + \frac{1}{2}P\{N_A > N_B = N_C\} + \frac{1}{6}P\{N_A = N_B = N_C\}. \end{aligned} \quad (6)$$

Next, by permuting the arguments, $P(C2) = h(p, r, q)$, $P(C3) = h(q, p, r)$, $P(C4) = h(q, r, p)$, $P(C5) = h(r, p, q)$, $P(C6) = h(r, q, p)$.

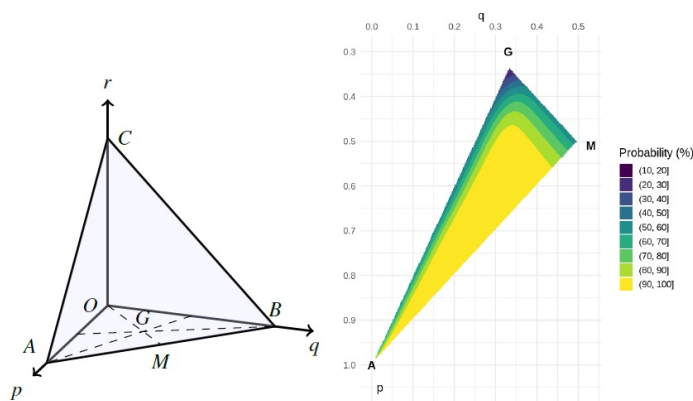
It suffices to calculate the probabilities of twelve or thirteen mutually exclusive and exhaustive subsets of the support set $\mathcal{S}$, shown in Figure 2(b), by sorting the elements of the subsets and then adding the trinomial($n, (p, q, r)$) probabilities of the points within each subset. Thereafter, we compute $P(C1)$, etc. using (6).

Sort the elements of the subsets efficiently, then add the probabilities.

Clearly, $P\{S_0\} = P\{N_A = N_B = N_C\} = \binom{n}{m, m, m} p^m q^m r^m$, only when $n = 3m$, and 0 otherwise. Next, letting $\lfloor u \rfloor$ denote the largest integer no more than u , we have

The central subset is a singleton if 3 divides n ; otherwise, it is empty.

Figure 3. The probability of correctly concluding “ $p > q > r$ ” inside the triangular region AGM where $\{p > q > r\}$.



(a) The trinomial parameter space partitioned by ordering the parameters $\triangle AGM$ where $\{p > q > r\}$. (b) For $n = 100$, contour plot of $P(\text{conclude “}p > q > r\text{”})$ inside $\triangle AGM$ where $\{p > q > r\}$.

$$\begin{aligned}
 P\{S_3\} &= P\{N_A > N_B = N_C\} = \sum_{\substack{x=\lfloor (n+2)/3 \rfloor \\ (n-x) \text{ even}}}^n f_n\left(x, \frac{n-x}{2}, \frac{n-x}{2}\right); \\
 P\{S_2\} &= P\{N_A = N_B > N_C\} = \sum_{\substack{z=0 \\ (n-z) \text{ even}}}^{\lfloor (n-2)/3 \rfloor} f_n\left(\frac{n-z}{2}, \frac{n-z}{2}, z\right); \\
 P\{S_1\} &= P\{N_A > N_B > N_C\} = \sum_{z=0}^{\lfloor n/3 \rfloor - 1} \sum_{y=z+1}^{\lfloor (n-z-1)/2 \rfloor} f_n(n-y-z, y, z).
 \end{aligned}
 \tag{7}$$

Caution: In Figure 3(b), the vertical axis is reversed to match with the 3D Figure 3(a); hence, the horizontal axis is shown on top. You can extend Figure 3(b) to the whole parameter space by successively reflecting on edges GA and GM and on their reflections!

Figure 3 shows the contour color plot of the probability of correctly concluding “ $p > q > r$ ” for $n = 100$, over the domain

$$\{(p, q) : p > q > r = 1 - p - q > 0\}.$$

This probability is the same at all six permutations of the parameters, as can be verified by renaming the three events. The R codes are omitted from this paper. Interested readers may email the first author to request access to the codes.

For each n , the probability of concluding that “ $p > q > r$ ” exhibits the following trends:

n	1500	1875	2203	2209	2210
$P("p > q > r")$	.89657	.93008	.94967	.94996	.95001

For each fixed $q \in (0, 1/2)$, $P("p > q > r")$ is a monotonically increasing function of $p \in (1 - 2q, 1 - q)$. For each fixed $p > 1/3$, $P("p > q > r")$ increases over the interval $((1 - p)/2, 1/3)$ and then decreases as q varies over $(1/3, p)$.

Conversely, when the true probabilities are $(p, q, r) = (.37, .33, .30)$, how large a sample must we take so that with probability at least 95% we will make the correct conclusion that " $p > q > r$ "?

The computations in *Table 2*, justify that the requisite n must be at least 2210. For each fixed (p, q, r) in ΔAGM , $P("p > q > r")$ increases as n increases. As shown in *Table 1*, we can first search for n over a large interval with large increments, then successively zoom in. Alternatively, we can automate the discovery of n using a binary search algorithm. See R codes in the *Appendix*. Starting with a wide interval of possible values for n , at each step, we check if $P("p > q > r")$ at the midpoint of the current interval exceeds the desired probability. If it does, we move to the left (lower) half-interval; if not, we move to the right (upper) half-interval. This binary search algorithm continues until we find the smallest n that yields the requisite probability. However, when n is large (over 170), calculating factorials becomes impossible. Therefore, we work with their logarithms.

We leave it to the reader to check that the answer to the same question when $(p, q, r) = (.35, .33, .32)$ is at least 17,715.

Table 2. The probability of correctly concluding that " $p > q > r$ " when $(p, q, r) = (.37, .33, .30)$ for various values of n .

We can use permutation symmetry to extend *Figure 3* onto $\{(p, q) : p, q, r = 1 - p - q > 0\}$.

Readers, please check these trends.

Determine sample size to achieve high enough accuracy.

A binary search algorithm discovers n .

Transforming to logarithms keeps the computations feasible.

Run the R codes after assigning $p = .35$ and $\text{right} = 20000$.

5. Identifying the Most Likely Event

For the dichotomous random phenomenon, ranking the two events is equivalent to identifying the more likely event. However, for the trichotomous random phenomenon, identifying the most probable event is a simpler task because you do not have to rank the other two events correctly.

N_A must be the largest. Then either N_A exceeds both N_B and N_C , or N_A equals one of N_B and N_C , or all three frequencies are tied.

When do we declare A is the most likely event among $\{A, B, C\}$? The following possibilities are permitted: $\{N_A > N_B > N_C\}$; or $\{N_A > N_C > N_B\}$; or $\{N_A > N_B = N_C\}$. Also, if $\{N_A = N_B > N_C\}$ or $\{N_A = N_C > N_B\}$, we can roll a fair die and declare A is the most likely event if an odd number appears on the die. Finally, if $\{N_A = N_B = N_C\}$, we can roll a fair die and declare A is the most likely event if a one or a four appears on the die. Therefore, the probability of declaring A as the most likely event is

$$\begin{aligned} g_n(p, q, r) &= P\{N_A > N_B > N_C\} + P\{N_A > N_C > N_B\} \\ &\quad + P\{N_A > N_B = N_C\} + \frac{1}{2}P\{N_A = N_B > N_C\} \\ &\quad + \frac{1}{2}P\{N_A = N_C > N_B\} + \frac{1}{3}P\{N_A = N_B = N_C\} \\ &= P(C1) + P(C2) = h(p, q, r) + h(p, r, q). \end{aligned} \quad (8)$$

In case of a two-way tie, A is declared as the most likely with probability $1/2$; in case of a three-way tie, with probability $1/3$.

All these expressions follow by symmetry.

By interchanging the roles of B and C , we note that $g_n(p, q, r) = g_n(p, r, q)$.

Similarly, we declare B is the most likely event with probability $g_n(q, r, p) = h(q, r, p) + h(q, p, r)$, and C is the most likely event with probability $g_n(r, p, q) = h(r, p, q, r) + h(r, q, p)$.

6. Summary and Extension

We studied the probability of correctly ranking the outcomes of a random phenomenon with two or three mutually exclusive and exhaustive events based on a random sample. We must efficiently sum the PMF's of a binomial or a trinomial distribution.

We invite interested readers to solve the problem of ranking the events of a random phenomenon that has four or more mutually exclusive and exhaustive events. Specifically, they may advise the geology club about the requisite sample size.

For instance, if the five mineral types (after sorting) have relative abundance (.15, .18, .20, .22, .25), then $n = 245$ rock specimens

We studied the problem of classifying into two or three categories.

Readers, may extend the solutions to classifying events into four or more categories.



give only a 22% chance of correct ranking. Likewise, $n = 2900$ specimens give a 90% chance of correct ranking. To achieve a 95% chance of correct ranking, we need $n = 3900$ specimens. These results are based on a simulation using the R codes given in the *Appendix*. Readers may proceed to derive the theoretical results. They may also verify the theoretical results we have derived for two or three categories via simulation.

Simulation is a good guide to discovering (and verifying) theoretical results.

Interested readers wishing to pursue the ranking problem further may adopt a hierarchical Bayesian approach that utilizes available reliable prior information on the relative ranking of the events, perhaps from previous studies. This would reduce the current sample size required to attain a desired level of precision.

A Bayesian approach uses reliable prior information to reduce the sample size.

For example, in the two-event problem, a beta prior on $P(A)$, with a binomial sampling distribution, will lead to a beta posterior on $P(A)$. (The beta distribution is the conjugate class for binomial samples.) A Bayesian analysis with a non-informative prior distribution (in this example, a uniform(0, 1) or a beta(1, 1) prior) should roughly agree with our classical analysis. For example, if $p = .55$, a Bayesian analysis with a beta(1, 1) prior on $P(A)$ concludes that the posterior probability of $\{P(A) > .5\}$ exceeds 95% when $n = 272$. See R codes in the *Appendix*. Therefore, if in a previous sample, we found 55 male fish and 45 female fish (based on which we agreed $p = .55$), then we need 172 additional fish in the current sample. On the other hand, if $p = .52$, then we need 1592 fish in the current sample for a total of $n = 1692$ fish.

The classical analysis is a special case of Bayesian analysis with a non-informative prior.

Similarly, in higher category ranking problems, a Dirichlet prior distribution on the initial probabilities of the multinomial sampling distribution yields a Dirichlet posterior distribution. We leave the details to the interested reader.

Acknowledgments

The authors never met in person. This collaborative research was conducted via Zoom meetings. The authors thank their respective institutions for providing the necessary tools. The authors also thank a referee for suggesting the Bayesian approach.



Suggested Reading

- [1] F S Amira, M M Rahman, B Y Kamaruzzaman, K C A Jalal, M Y Hossain, N S Khan, Relative abundance and growth of male and female *Nemipterus furcosus* population, *Sains Malaysiana*, 45(1), pp.79–86, 2016.
- [2] Steven Earle, *Physical Geology*, 2 Edition. British Columbia Ministry of Advanced Education and Skills Training, and the Hewlett Foundation, 2019.
- [3] Wikipedia, the Free Encyclopedia. Law of large numbers.
https://en.wikipedia.org/wiki/Law_of_large_numbers
- [4] Vijay K. Rohatgi and A. K. Md. Ehsanes Saleh, *An Introduction to Probability and Statistics*, John Wiley & Sons, Wiley Series in Probability and Statistics, ISBN 9781118799642, 2015. DOI: 10.1002/9781118799635.
- [5] Wikipedia, the Free Encyclopedia. Recurrence relation.
https://en.wikipedia.org/wiki/Recurrence_relation
- [6] Wikipedia, the Free Encyclopedia. Pascal's triangle.
https://en.wikipedia.org/wiki/Pascal's_triangle
- [7] Ronald Walpole, Raymond Myers, Sharon Myers and Keying Ye, *Probability & Statistics for Engineers and Scientists*, 9 ed. Boston, MA: Pearson, 2017.
- [8] R Core Team, R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria, 2021. <https://www.R-project.org/>.

APPENDIX

Below are the R codes to draw *Figure 1*, to compute *Tables 1* and *2*, and to simulate ranking 5 events. The reader can access the R codes to draw *Figure 3* by emailing the first author.

```
# Figure 1
p_vals <- seq(0.3, 0.7, length.out = 1001)
n_vals <- c(50, 100, 200, 400)
plot(NULL, xlim = c(0.3, 0.7), ylim = c(0, 1),
      xlab = "p", ylab = "", las = 1)
mtext('P("p > q")', side=2, line=0, las=1, padj=-24)
for (n in n_vals) {
  prob <- sapply(p_vals, function(p)
    {1-pbinom(n/2, n, p) + 0.5*dbinom(n/2, n, p)})
  lines(p_vals, prob, lty=which(n_vals == n), lwd=2) }
legend("bottomright", legend=paste("n =", n_vals), lwd=2)
```



CLASSROOM

```
# Table 1: P(C)=P{conclude "p>q"} as a function of n
p=.55 # .52
pc=function(n){1-pbinom(n/2,n,p) + dbinom(n/2,n,p)/2}
for (n in seq(100,300,50)){print(pc(n))}
for (n in seq(255,295,10)){print(pc(n))}
for (n in 266:270){print(pc(n))}
for (n in seq(1200,2000,200)){print(pc(n))}
for (n in seq(1610,1790,20)){print(pc(n))}
for (n in seq(1687,1695,2)){print(pc(n))}

# Table 2: Given n compute P{conclude "p>q>r"}
# Determine n so that P{conclude "p>q>r"} >= threshold
# Parameters
p <- 0.37 # 0.35
q <- 0.33
r <- 1 - p - q
threshold <- 0.95

# Safe multinomial probability computation
safeComputeP <- function(n, p, q, r) {
  log_p <- log(p)
  log_q <- log(q)
  log_r <- log(r)
  log_n_fact <- lgamma(n + 1)
  # initialize the probabilities of sets S0, S1, S2, S3
  total0 <- 0 # P{N_A = N_B = N_C}
  total1 <- 0 # P{N_A > N_B > N_C}
  total2 <- 0 # P{N_A = N_B > N_C}
  total3 <- 0 # P{N_A > N_B = N_C}

  # Case S0: N_A = N_B = N_C
  if (n %% 3 == 0) {
    m <- n/3
    log_term <- log_n_fact - 3*lgamma(m+1) +
      m*(log_p + log_q + log_r)
    total0 <- exp(log_term) }

  # Case S1: N_A > N_B > N_C
  for (z in 0:(floor(n/3)-1)) {
    y_min <- z + 1
    y_max <- floor((n-z-1)/2)
    if (y_min > y_max) next
    for (y in y_min:y_max) {
      x <- n - y - z
      log_term <- log_n_fact - lgamma(x+1) - lgamma(y+1) -
        lgamma(z+1) + x*log_p + y*log_q + z*log_r
      total1 <- total1 + exp(log_term)
    }
  }
}
```

CLASSROOM

```
# Case S2: N_A = N_B > N_C
for (z in 0:floor((n-2)/3)) {
  x <- y <- floor((n-z)/2)
  if (2*x+z != n) next
  log_term <- log_n_fact -lgamma(x+1) -lgamma(y+1) -
    lgamma(z+1) +x*log_p +y*log_q +z*log_r
  total2 <- total2 + exp(log_term)
}

# Case S3: N_A > N_B = N_C
for (x in (floor((n+2)/3)):n) {
  yz <- (n-x)
  if (yz %% 2 != 0) next
  y <- z <- yz/2
  log_term <- log_n_fact -lgamma(x+1) -lgamma(y+1) -
    lgamma(z+1) +x*log_p +y*log_q +z*log_r
  total3 <- total3 + exp(log_term)
}

return(total1 + 0.5*total2 + 0.5*total3 + (1/6)*total0)
}

cat("Full Binary Search:\n")
left <- 0
right <- 3000 #20000
answer <- NA

while (left <= right) {
  mid <- floor((left + right)/2)
  prob <- safeComputeP(mid, p, q, r)
  cat(sprintf("n = %d    P = %.6f\n", mid, prob))

  if (prob >= threshold) {
    answer <- mid
    right <- mid-1}
  else {left <- mid + 1}
}

cat(sprintf("\n Min n for P >= %.2f is: %d\n",
  threshold, answer))

### simulation for P(correct ranking of five minerals)
x=c(1,2,3,4,5)
pr=c(.15, .18, .20, .22, .25)
play=function(){
  ta=table(sample(x, 245, replace=T, pr)) # 2450, 2900, 3900
  (sum((ta==sort(ta)))==5)}
data=replicate(10^4, play())
table(data)/10^4 # prop(TRUE) estimates P(correct ranking)
```



CLASSROOM

```
### Bayesian analysis with non-informative beta(1, 1) prior
n=seq(251, 300, 1) # proposed values of n
p=.55; r=n*p      # expected frequency of A
1-pbeta(.5, r+1, n-r+1) # posterior probability {P(A)>.5}
```

